Setup and Dataset

Q1. Import the necessary Python libraries (Pandas and NumPy).

Q2. Create a dataset named df with the following columns:

Student_ID	Height_cm	Weight_kg	Hours_Studied	Attendance	Assignment_Score	Final_Grade
1	160	55	4	80	75	B
2	170	68	6	90	85	A
3	165	60	5	85	78	B
4	180	85	7	95	92	A
5	175	78	3	70	65	C

Creating New Features (Feature Derivation)

Q3. Create a new column Height_m by converting Height_cm to meters. (Hint: divide by 100)

Q4. Derive a new feature `BMI` (Body Mass Index):

$$BMI = \frac{mass}{height^2}$$

Add this as a new column to the DataFrame.

Q5. Create another feature

`Study_Efficiency = Hours_Studied × Attendance.`

Q6. Create,

`Performance_Ratio = Assignment_Score / Hours_Studied`.

Feature Selection

Q7. Identify which columns are original and which are derived.

(Hint: Make a Markdown list with both categories.)

Q8. Drop the column `Height_m` (it was only used for calculating BMI).

Q9. Display the correlation matrix for all numerical features.

(Hint: use `df.corr(numeric_only=True)`)

Q10. Which derived features appear to have a strong correlation with others?

Write your observation in a Markdown cell.

Feature Importance (Conceptual)

Q11. Why is feature engineering important before model building?

(Hint: Helps create more informative data for machine learning.)

Optional Advanced Tasks (For Extra Practice)

Q12. Feature Encoding

Convert the `Final_Grade` column into numeric values:
A → 3, B → 2, C → 1
Add a new column `Final_Grade_Code`.

Q13. Feature Scaling on Derived Feature

Normalize the `Study_Efficiency` column using MinMaxScaler so that values lie between 0 and 1.
Add a new column `Study_Efficiency_Scaled`.

Q14. Visual Correlation Analysis

Create a heatmap to visualize correlation among all numerical columns.
(Hint: use `seaborn.heatmap()`)

Q15. Outlier Detection in Engineered Feature

Find if any students have BMI > 30 (possible outliers).

Reflection (Markdown)

Q16. Which engineered feature best represents student performance and why?

Q17. Why should we drop irrelevant or redundant features?

Q18. If you were to build a prediction model, which 3 features would you select and why?